# Hang Gao

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## Education

**Rutgers University-New Brunswick** — Piscataway, NJ, USA
Ph.D. student in Computer Science — September 2022-Present
**Syracuse University** — Syracuse, NY, USA
Master of Computer Science — September 2020-May 2022
**Jilin University** — Changchun, China
B.Eng. in Logistics Engineering — September 2016-July 2020

## Experience

**Syracuse University, Syracuse, NY** — **Research Assistant** — June 2021-May 2022
*Fairness in Social Network.*
- Revolutionized social network analysis by developing equitable sparsification methodologies, enhancing data interpretability and maintaining structural integrity in graph representations.
- Reduced community expansion discrepancies by 18% through an optimized edge-weight algorithm, improving the accuracy of edge-deletion.
- Pioneered the 'Redundancy Edge' concept to refine the Scale-free network structures, demonstrating its effectiveness on a complex network via recursive edge elimination.

**Rutgers University, New Brunswick, NJ** — **Research Assistant** — September 2023-Present
*Reliable Retrieval and Reasoning for Knowledge-Intensive AI. (Advisor: Dimitris N. Metaxas)*
- Designed LLM agent systems for more reliable and efficient task execution, including tool recommendation and shared memory mechanisms that improve tool selection, cross-agent knowledge reuse, and response quality in complex tasks.
- Developed retrieval and reasoning frameworks for text and knowledge graphs, including parameter-free retrieval balancing relevance and diversity, and robust graph reasoning that recovers hidden semantic links in noisy or sparse data.
- Built multimodal RAG methods that improve evidence selection and confidence-controlled answer generation, enabling more accurate and trustworthy visual question answering without model retraining.

**NEC Laboratories America, Princeton, NJ** — **Research Intern** — May 2025-August 2025
*Efficient GraphRAG for Domain-Specific QA. (Mentors: Yangmin Ding & Shaobo Han)*
- Designed a multi-layer bipartite graph to enable efficient multi-hop reasoning in domain-specific RAG.
- Reduced graph construction cost and eliminated entity–entity hallucinations by avoiding explicit entity alignment, improving robustness in real-world deployment.
- Enhanced update efficiency and adaptability of RAG systems, supporting dynamic knowledge integration.

## Selected Publications

[1] **Hang Gao**, Wujiang Xu, Kai Mei, Dimitris N. Metaxas. "Semantic Shift: the Fundamental Challenge in Text Embedding and Retrieval".

[2] **Hang Gao**, Wujiang Xu, Zhixing Zhang, Kai Mei, Jingyi Yang, Dimitris N. Metaxas. "CLIMB: Confidence-Guided Complementary Evidence for Multimodal Retrieval-Augmented Generation".

[3] **Hang Gao**, Dimitris N. Metaxas. "Beyond Explicit Edges: Robust Reasoning over Noisy and Sparse Knowledge Graphs". (ICML 2026)

[4] **Hang Gao,** Yongfeng Zhang. "Task-Aligned Tool Recommendation for Large Language Models".(IJCNLP-AACL 2025 (Oral))

[5] Wujiang Xu, Zujie Liang, Kai Mei, **Hang Gao**, Juntao Tan, Yongfeng Zhang. "A-mem: Agentic memory for llm agents". (Neurips2025)

[6] **Hang Gao**, Dong Deng, Yongfeng Zhang. "Vector Retrieval with Similarity and Diversity: How Hard Is It?".

[7] **Hang Gao,** Yongfeng Zhang. "INMS: Memory Sharing for Large Language Model based Agents".

## Teaching

2024 SP CS 214 – Systems Programming
2023 FA CS 512 – Introduction to Data Structures and Algorithms
2023 SP CS 509 – Foundations of Computer Science
2022 FA CS 416 – Operating Systems Design

## Technical Skills

- **Programming Skills**: Python(expertise), Java, C++
- **Frameworks**: Pytorch, Networkx

## Academic Services

**Conference Reviewer:**
- International Conference on Learning Representations (ICLR)
- International Conference on Machine Learning (ICML)
- Neural Information Processing Systems (NeurIPS)
- European Conference on Computer Vision (ECCV)
- Association for the Advancement of Artificial Intelligence(AAAI)
- Conference on Language Modeling(COLM)